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Department of Computer Science and Engineering

B Tech. Computer Science & Engineering
Semester – 2
23CSE111 OBJECT ORIENTED PROGRAMMING

Guidelines for Object-Oriented Programming (OOP) Project (Jan -June 2025)

The objective of this academic project is to enable students to learn and apply Object-Oriented Programming (OOP) concepts such as encapsulation, inheritance, polymorphism, and abstraction in solving real-world problems. Students will design and implement a structured, modular, and reusable software solution using OOP principles while enhancing their problem-solving and logical thinking skills. A key aspect of the project is system design using Unified Modelling Language (UML), where students will create class diagrams, use case diagrams, and sequence diagrams to plan their software before coding.

1. Team Selections

Each team may contain minimum of 4 members.
Every member must actively contribute to the project.
Every member must explain their contribution during the final project demo & viva.
Marks may be deducted if contributions are found to be unequal.

2. Project Topic Selection

Choose a project topic that effectively demonstrates Object-Oriented Programming principles such as Encapsulation, Inheritance, Polymorphism, and Abstraction.
The topic should be relevant, innovative, and feasible within the given timeframe.
Avoid overly complex ideas that may not be completed within the given duration.
If students select common topics, they will be evaluated together, irrespective of their batch.
Sample domains include:

- Library Management System
- Hospital Management System
- Online Banking System
- E-commerce Application
- Student Management System
- Employee Payroll System

3. Project Expectations

Each project should demonstrate:

- Proper Class Design: Use well-structured classes with appropriate attributes and methods.
- OOP Concepts: Implement encapsulation, inheritance, polymorphism, and abstraction in meaningful ways.
- Code Modularity: Organize the code into separate files/modules for maintainability.
- User Interaction: Provide a simple and functional user interface for interactions.
- Database Connectivity (if applicable): Use proper data management techniques.
- Error Handling: Implement exception handling and validation to avoid system crashes.
- Code Documentation: Write meaningful comments and maintain good coding practices.

4. Project Requirements & Expectations

(a) System Design Using UML

Students must create UML diagrams for project planning, including:

- Class Diagram – Representing the structure of the system.
- Use Case Diagram – Mapping out interactions between users and the system.
- Sequence Diagram – Illustrating the flow of processes within the system

(b) Front-End & Back-End Requirements

- Front-End: GUI-based using Java Swing, JavaFX, or HTML/CSS for web-based projects.
- Back-End: The core logic should be implemented using Java.
- Database (if applicable): Projects requiring data storage should use MySQL, SQLite, or JSON/CSV for file-based storage.

5. Evaluation Criteria

- UML Design (25%): System design using UML diagrams.
- Application of OOP (25%): Apply OOP concepts effectively to solve the problem.
- Code Quality & Documentation (10%): Readability, comments, and proper structure.
- User Interface (5%): Ease of use and interactive design.
- Demo & Viva (35%): Ability to explain the project, coding approach, and OOP concepts.

6. Submission & Deadline

Submit source code along with a project report containing:

- Introduction & Objectives
- System Design (Class Diagrams, Use Case Diagrams)
- Implementation Details
- Screenshots of Execution
- Conclusion & Future Scope

Deadline for submission:

- System Design - 06.05.2025
- Demo & Viva – 29.05.2025

7. Common Topic Evaluation

If multiple batches choose the same topic, they will be evaluated together.
Groups should ensure uniqueness in their approach, design, and implementation.